GlucoTwin — system architecture

A patient-specific type 2 diabetes digital twin: a static electronic health record fused with continuous glucose, wearable and meal streams, producing calibrated forecasts, adverse-event probabilities and counterfactual “what if” answers for a clinician.

Digital Twin Challenge 2026
Indian Institute of Technology Guwahati
Python 3.11 · LightGBM · PyTorch · Streamlit
Runs on a laptop CPU · MIT licence

1 · DATA SOURCES — ANONYMISED, OPEN OR SYNTHETIC

REAL · OPEN
CGMacros (PhysioNet)
45 adults. CGM, Fitbit heart rate and activity, meal macros, labs. Main development cohort.

REAL · OPEN
Shanghai T1DM / T2DM
100 T2DM patients. CGM, meals, labs. Second cohort for cross-population transfer.

SIMULATED
Synthetic Indian cohort
CGM, heart rate, steps, sleep stages, HRV, IFCT-2017 Indian meals, lowered insulin sensitivity.

SIMULATED
Static EHR
Demographics, diagnoses, medications, eGFR, HbA1c, genetic-marker dosages. Synthea-compatible schema.

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2 · HARMONISATION AND FEATURE FUSION `SRC/DATA` · `SRC/FEATURES`

Common grid
Every cohort resampled to one 5-minute timeline with a shared schema and verified checksums.

Causal features only
No feature may use information from after the prediction instant. Enforced in code and tested.

Patient-grouped splits
Train, inner-validation, calibration and holdout patients never overlap.

Four streams fused
Recent CGM · time of day · meal macros · heart rate and activity · static EHR profile.

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3 · MODEL LAYER

Fusion forecaster
Tuned LightGBM on the change from the current value, at +30, +60 and +120 min. The served model.

Mechanistic twin
Minimal glucose–insulin ODE fitted per patient. RK4 integration, extended Kalman filter state.

Personalisation
Population model plus fine-tuning or recursive least squares on the patient's own first days.

Federated simulation
FedAvg / FedProx across simulated clinics. Model updates are shared; records never are.

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4 · UNCERTAINTY AND CALIBRATION `SRC/UNCERTAINTY`

80 % prediction intervals
Split-conformal and CV+. Measured coverage 80.8 % (CGMacros) and 79.5 % (Shanghai) at +60 min.

Adverse-event probabilities
Platt / isotonic calibration for crossing 180, 140 and 70 mg/dL. Reported as probabilities, not labels.

Alert policy
Alert burden curves: episodes caught per false alarm per day, so a clinic can choose its threshold.

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5 · CLINICAL OUTPUTS

Forecast + band
30 / 60 / 120-minute trajectory with an honest 80 % interval and one plain-language sentence.

Adverse-event risk
Calibrated chance of hyperglycaemia or hypoglycaemia in the window.

What-if engine
Counterfactual meal, carbohydrate, timing and post-meal walk, through a monotone model and the twin.

Trust surface
Data quality, personalisation depth, SHAP attribution, and the limits of each answer beside it.

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6 · CLINICIAN INTERFACES

Streamlit dashboard `make app`
Ten pages: Overview, Live twin, Glucose report (AGP, time in range, GMI), What-if lab, Twin state, Personalization, Input tiers, Data quality, Evidence, PDF export. mg/dL ⇄ mmol/L switch.

Web console `web/`
Single-page clinician console: patient list with triage, forecast, what-if lab, glucose report, health record, sleep and activity context, visit summary, and a role-gated access log.

7 · REPRODUCIBILITY AND ENGINEERING

One command per stage
`make data` · `train` · `calibrate` · `app-data` · `web-real`. Every figure and table regenerates from source.

Continuous integration
GitHub Actions on Python 3.11: ruff, black, pytest and an end-to-end smoke run on every push.

Pinned and containerised
Pinned dependencies and a Docker image, verified on a clean machine.

Documented data
Datasheet and model card, checksum-verified loaders, and a written limitation for every claim.

GOVERNANCE — THE METHOD

- **Pre-registered decisions.** Every protocol and decision rule is written to `docs/DECISIONS.md` before the run that tests it. 70+ entries.
- **Paired patient bootstrap.** Every comparison carries a confidence interval over patients, not windows.
- **Untouched holdout.** Final claims are confirmed on patients never used for any choice.
- **Null results reported.** Where a signal does not help, the number is published rather than dropped.

PRIVACY BY DESIGN

- Only anonymised open or synthetic data. No real patient record enters the repository.
- **Federated path:** raw records stay inside the clinic; only model updates are exchanged.
- Role-gated clinician access with a session access log.
- Handling mapped to India's **DPDP Act** in `docs/limitations.md`.

HEADLINE RESULTS

11.6	16.8
MAE MG/DL · +30 MIN	MAE MG/DL · +60 MIN
0.82	**80.8 %**
AUROC · >180 MG/DL	INTERVAL COVERAGE

- **Transfer:** a model moved to a new population is 3.4 mg/dL behind a locally trained one; three days of the new patient's data close much of the gap.
- **Federated:** FedAvg matches pooled training (difference 0.09–0.12, intervals include 0) and beats a single small clinic by ≈2 mg/dL.

SCOPE AND LIMITS

- One condition: **type 2 diabetes**.
- Decision support only. Never recommends a dose or a drug. Not a medical device.
- India, sleep and federated results are **simulations**: they test design, not clinical performance.
- Not prospectively validated.

Code MIT. Datasets keep their own licences. Simulated streams labelled SIMULATED throughout.